

A liquid-phase first-order irreversible reaction



is carried out isothermally at 300 K in a well-mixed semi-batch reactor.

Rate constant $k = 0.4 \text{ min}^{-1}$ at 300 K

Initially, the reactor is filled with 20 moles of a solvent (inert species) with volume 2 dm^3 . At $t = 0$ a liquid stream containing the solvent and species A at concentration $C_{A0} = 0.05 \frac{\text{mol}}{\text{dm}^3}$ is fed to the reactor. The volumetric flow rate of the inlet stream is gradually reduced according to the equation $v_0(t) = v_{0i} e^{-\alpha t}$, where $v_{0i} = 2 \frac{\text{dm}^3}{\text{min}}$ and $\alpha = 1 \frac{1}{\text{min}}$. The molar ratio of solvent to species A in the inlet stream is 10.

- (1) Write down two mole balances for the species A and B and derive two first-order differential equations for concentrations of A and B in the reactor as functions of time.
- (2) Write a Python code to solve the above system of ODEs and plot the concentrations of A and B in the reactor as functions of time.

Mole balance

$$r_A = -kC_A$$

For species A since $N_A = C_A \cdot V$

$$\rightarrow \frac{dN_A}{dt} = r_A \cdot V + F_{Ai} \quad F_{Ai} = v_0(t) \cdot C_{A0}$$

$$\frac{dN_A}{dt} = \frac{d(C_A \cdot V)}{dt} = -kC_A \cdot V + v_0(t) \cdot C_{A0}$$

$$\underbrace{V \frac{dC_A}{dt}} + \underbrace{C_A \frac{dV}{dt}}_{\substack{\downarrow \\ v_0}} = -kC_A V + \underbrace{v_0(t) \cdot C_{A0}}_V$$

$$\frac{dC_A}{dt} + \frac{C_A}{V} v_0 = \frac{-kC_A V + v_0(t) \cdot C_{A0}}{V}$$

$$\frac{dC_A}{dt} = \frac{-kC_A V + v_0(t) \cdot C_{A0}}{V} - \frac{C_A v_0}{V}$$

$$\frac{dC_A}{dt} = -kC_A + \frac{v_0(t) \cdot C_{A0}}{V} - C_A \frac{v_0}{V}$$

$$\textcircled{1} \frac{dC_A}{dt} = \frac{v_0(t)}{V} (C_{A0} - C_A) - kC_A$$

For species B

No initial C_B

$$\frac{dN_B}{dt} = r_B V$$

$$r_B = +kC_A$$

$$V \frac{dC_B}{dt} + C_B \frac{dV}{dt} = r_B \cdot V$$

$$\cancel{V} \frac{dC_B}{dt} + C_B \frac{dV}{\cancel{V}} = \cancel{K C_B} \cdot \cancel{V}$$

$$\frac{dC_B}{dt} + \frac{C_B}{V} \frac{dV}{dt} = K C_B$$

$$\textcircled{2} \frac{dC_B}{dt} = K C_B - C_B \frac{V_0(t)}{V}$$

```
#Import all the library, without this, the code would not work
%matplotlib widget
import numpy as np #Array
from scipy.integrate import odeint #ODE solver
import matplotlib.pyplot as plt #To make the graphs
```

```
#Parameters
```

```
K = 0.4 #min-1
CA0 = 0.05 #mol/dm3
v0i = 2 #dm3/min
alpha = 1 #1/min
V0 = 2 #dm3 This is the volume where the reactor is filled at start, do not confundir with v0i (not the same!!!!)
```

```
#Volumetric flow (This problem is different from the others since volume changes with time)
```

```
def v0(t):
    return v0i * np.exp(-alpha * t)
```

```
#ODE part
```

```
def ODES(X, t):
    CA, CB, V = X
    vt = v0(t)

    dCAdt = (vt/V) * (CA0 - CA) - K*CA
    dCBdt = (K * CA) - (vt/V)*CB
    dVdt = vt

    return [dCAdt, dCBdt, dVdt]
```

```
#NOTE: Not sure if V is also a diff eq, but it seems that it doesnt change anything if we use it as diff eq or not :D
```

```
#Conditions
```

```
X0 = [0.05, 0, V0]
```

```
t = np.linspace(0, 10, 1000) #Time as an array always, in this case starts at 0 and adds 10 everytime until reaches 100
```

```
#SOLVE ODES
```

```
sol = odeint(ODES, X0, t)
CA = sol[:,0]
CB = sol[:,1]
V = sol[:,2]
```

```
#Plotting
```

```
plt.figure()
plt.plot(t, CA, label='CA')
plt.plot(t, CB, label='CB')
plt.xlabel('Time (min)')
plt.ylabel('Concentration (mol/dm3)')
plt.legend()
plt.grid()
plt.show()
```

